Application Layer Protocols :-

→ DNS, SMTP, POP, FTP, HTTP

DNS → will see.
SMTP, POP → email
FTP → files
HTTP → web pages.

DNS → Domain name ~~server~~

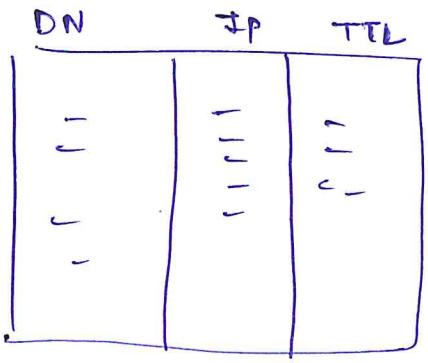
Google.com → Domain name

→ DNS helps us getting Ip address of the domain name.
→ DNS uses UDP protocol.
→ IP address are dynamic they can change.
→ If you do sudo command on terminal

nslookup google.com → Uses DNS protocol.

It will give you ip address of google.

DNS Tables :



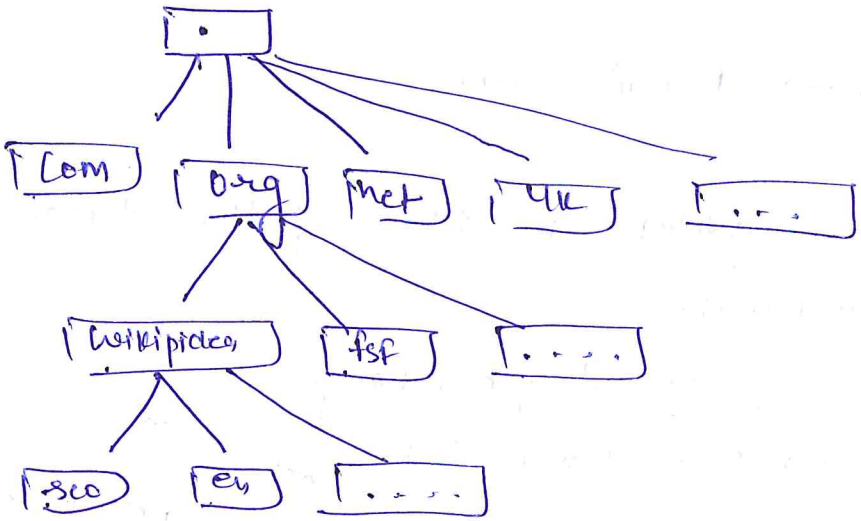
Domain names:

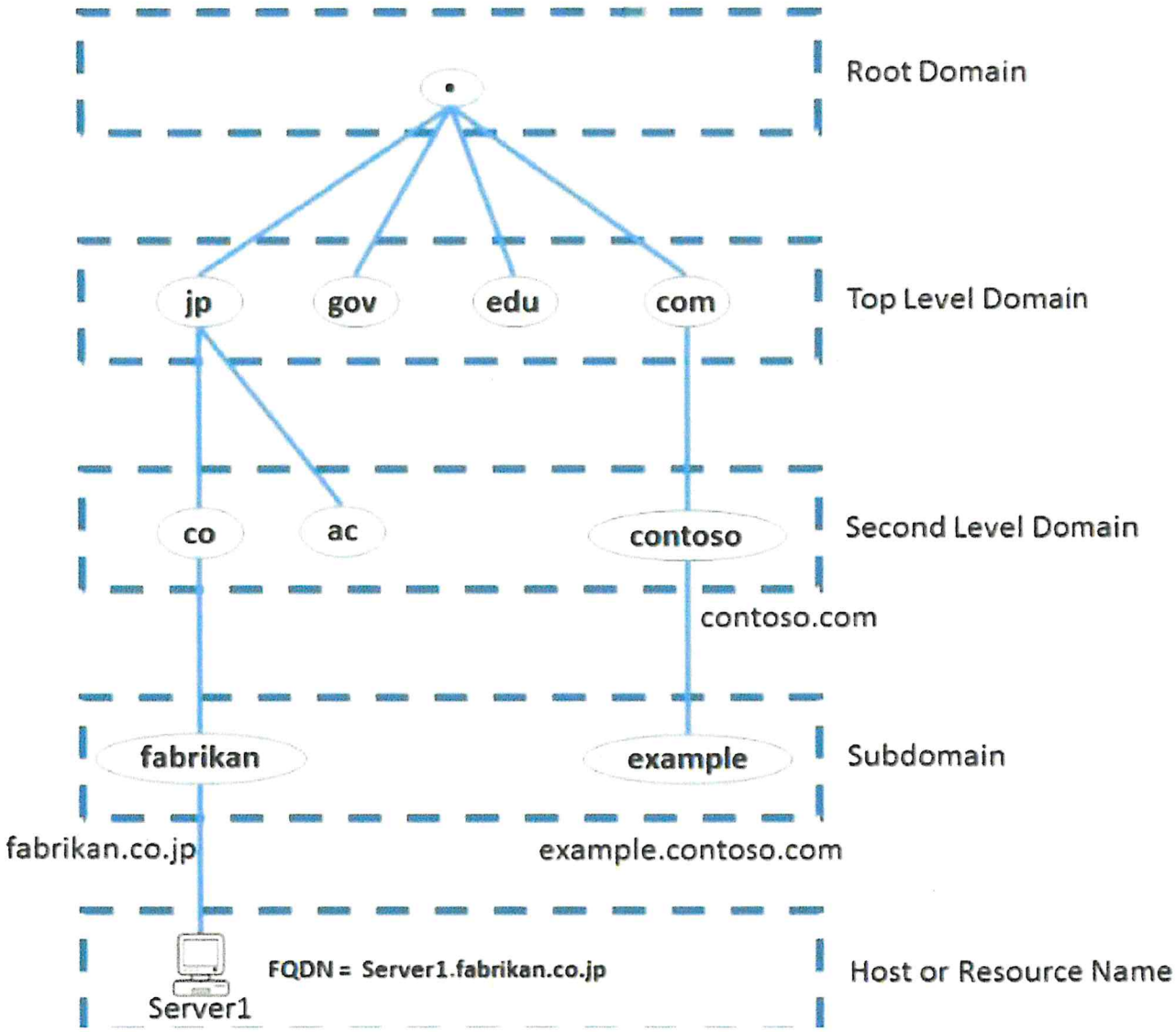
Generic domain: .com, .org, .edu, .net, .mil

Country domain: .in, .jp, .uk, .de

.in → India

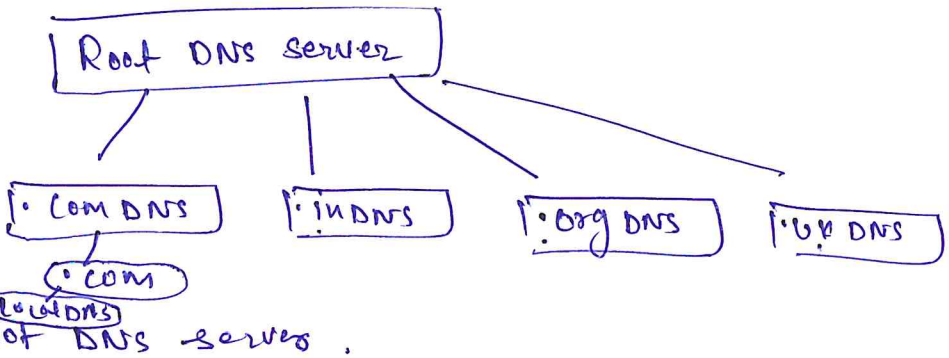
.jp → Japan





→ This is how the hierarchy of the DNS servers.

* DNS database organization :-

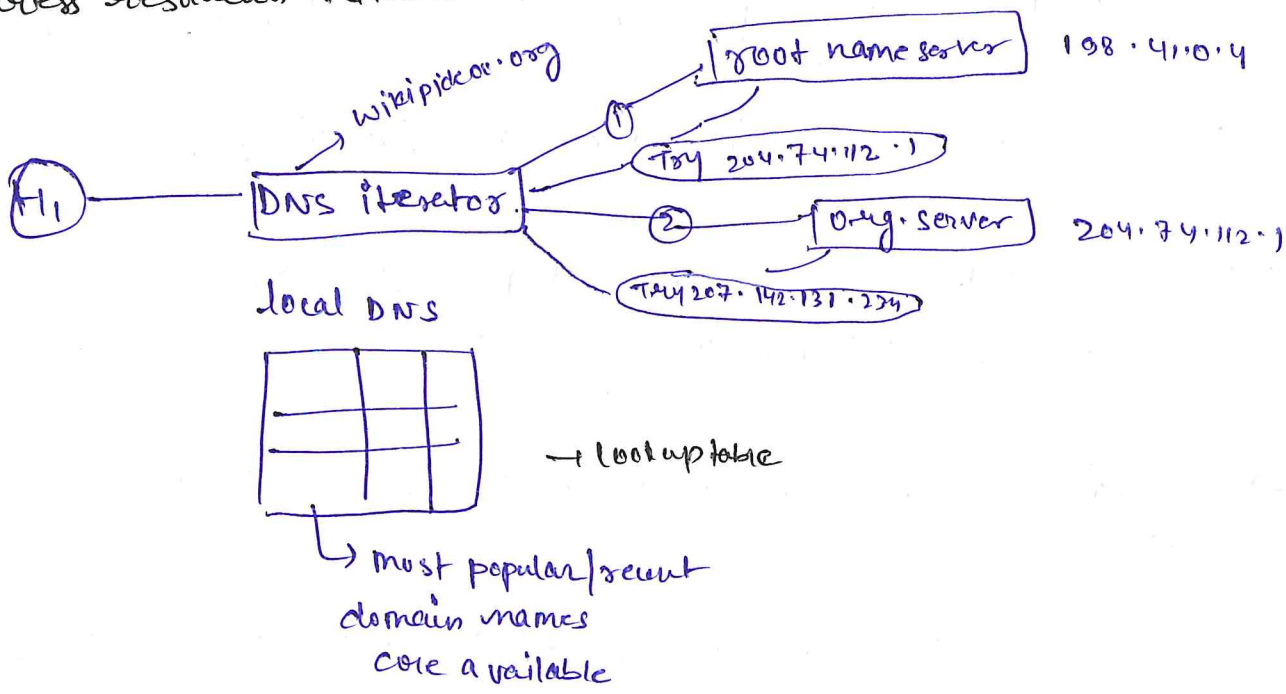


* We have 13 root DNS servers.

* Local DNS is ISP.

→ DNS entry table contains : Domain name, IP, validity

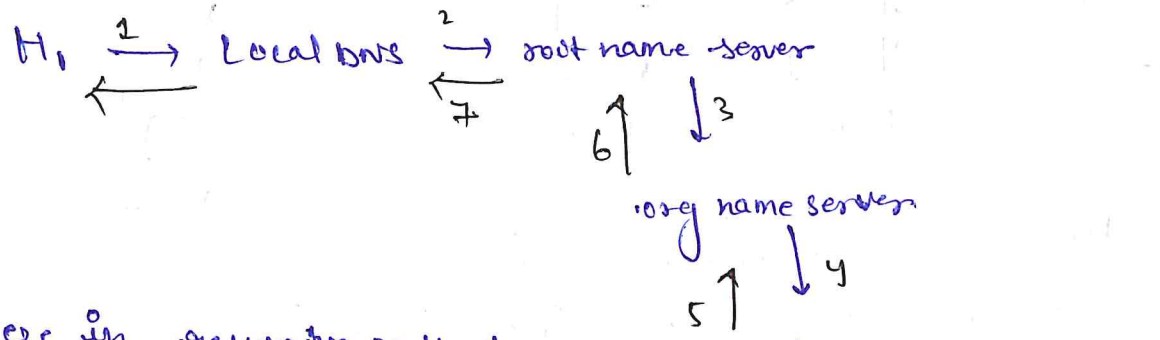
Address resolution : iterative method.



→ Let's take a case we are searching for wikipidea.org Ip address, this is not available in Local DNS, then request goes to root name server, It says I don't know please try 204.74.112.1. Now DNS req goes 204.74.112.1 (org server). It replies the answer.

Here we are using iterative method one by one.

Address resolution : Recursive



→ Here in recursive method the server forwarding the request to another server by itself. Here we are not doing iteratively.

Simple mail Transfer protocol and Post office protocol (POP)

→ not same as gmail (web-based email).

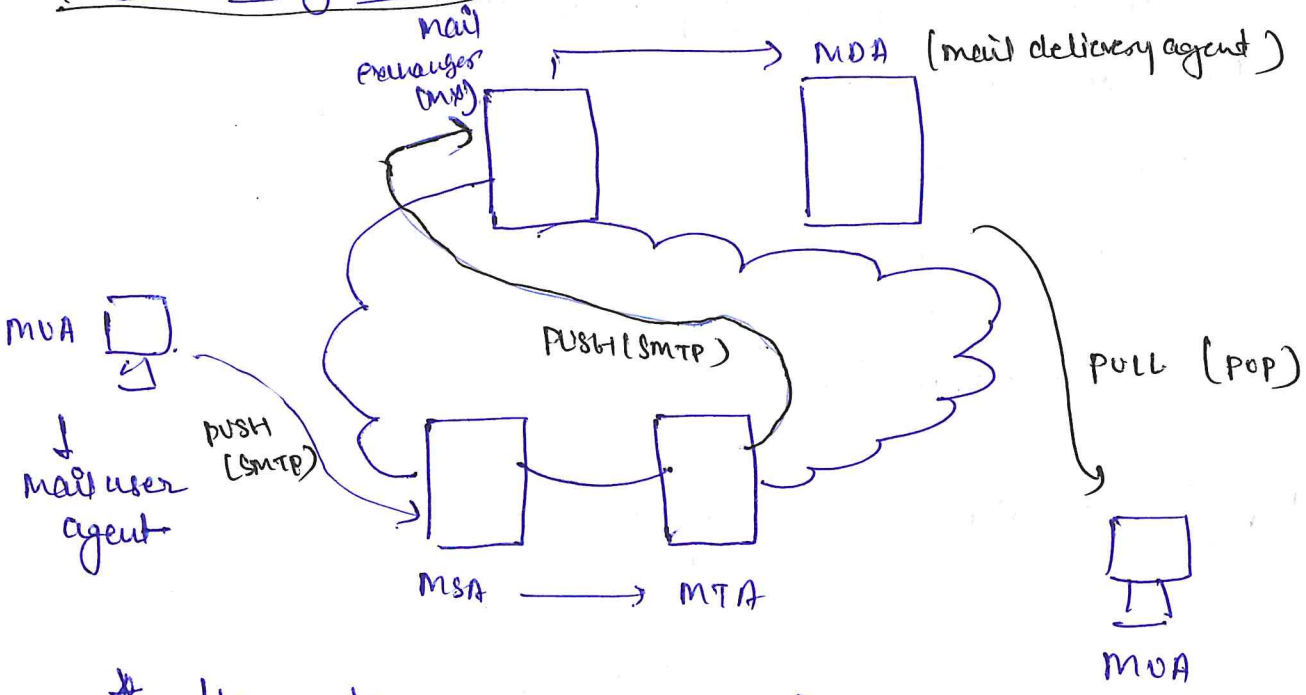
→ Ms outlook uses SMTP and POP for sending and receiving mail.

→ gmail: can use POP and SMTP to pull and push email.

↳ we can connect gmail in ms outlook to pull and push email.

* Inherently Gmail supports SMTP and POP.

Mail processing model:



* Here first we push the mail using SMTP from sender [mail address] to MSA.

* MSA is mail submission agent

→ MSA sends to MTA (mail transfer agent)

→ The mail transfer agent who are in local server of sender institution.

→ The mail transfer agent using DNS [mail exchanger record] gives us the address of email server of receiver cosporation.

→ The MTA pushes to mail exchanger

→ mail exchanger sends to MDA. [mail delivery agent]

→ The job of mail delivery agent saves the mail

→ The receiver MUA comes online, the MDA job is to pull the mail to the MUA.

→ Pop is used to pull the mail (post office protocol).

MIME → It helps us to send non-text data also like an attachment. (like image)

Multimedia
Internet
Mail extensions

Commands and data:

→ Port 25 for SMTP between mail servers (TCP or UDP) IANA.

→ In email commands and data on the same connection (in-band).

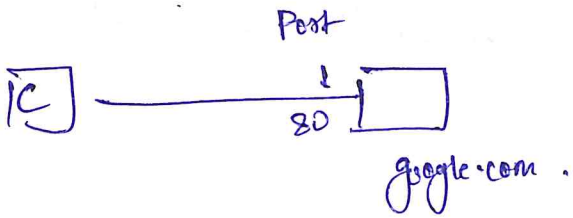
commands → PUSH & POP

data → email

Hypertext Transfer Protocol : (HTTP)

→ HTTP along with some security is HTTPS.

→ needs a reliable transport layer protocol (TCP)

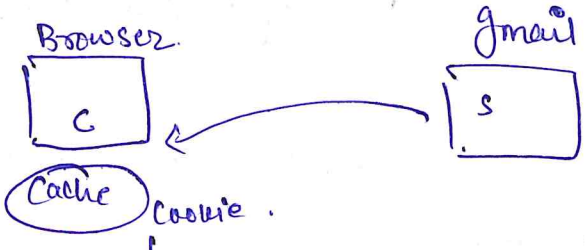


* HTTP uses port = 80

→ It is stateless protocol.

↳ The server doesn't maintain the state information.

→ Client maintains states using cookies. [in HTTP header]



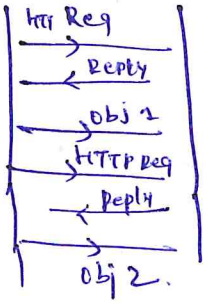
↓

All the information of state is stored in cookie like login information

→ HTTP 1.0 non-persistent connection

HTTP 1.1. Persistent connection

In Non-persistent → For every object you need to open a connection and close a connection



- Persistent - connection → For every object we need only one connection.

HTTP methods:-

GET → Get a webpage (w3c)

HEAD → Same as get but response header only

POST → filling the form [submit]

PUT → Upload an object in google drive.

DELETE → For deleting an object.

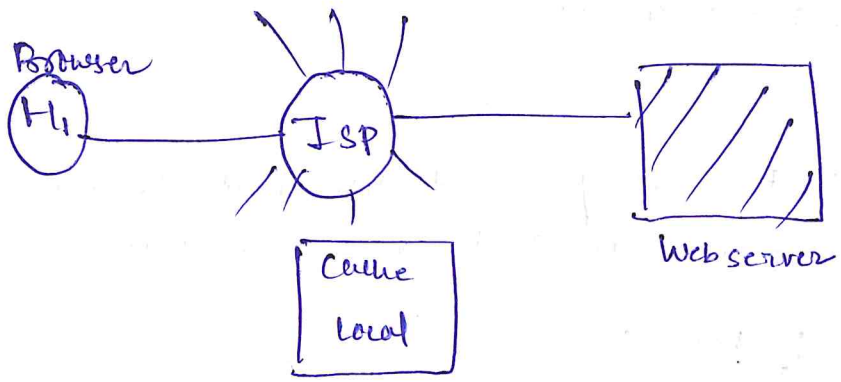
CONNECT → For connection of secure HTTPS

OPTIONS → what options are provided by server

TRACE → TRACE servers

PATCH → Partial modifications.

HTTP cache and TRACE



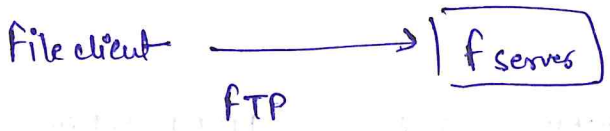
→ HTTP caching helps if some webserver page is locally available

→ In ISP we store some webpage that frequently not changing in cache local.

→ TRACE helps us to understand whether that page is served by ISP or webserver.

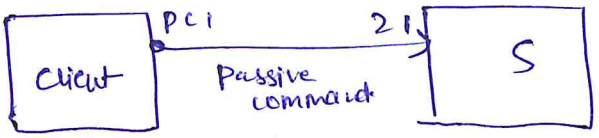
File Transfer Protocol (FTP)

This is used to transfer file from file server to a file client.

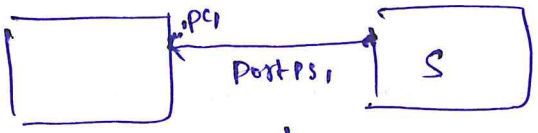


→ Uses reliable TCP

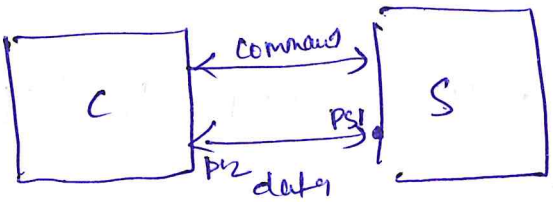
→ FTP is out of band. [different operation of command and data].



Ask for file



→ I will share the using PS1



→ FTP are stateful. They store all information of ~~state~~ in server log.

→ POP also uses stateful mechanism.

Que: Which of the following protocols pairs can be used to send and retrieve emails (in that order)?

(A) IMAP, POP3

(B) SMTP, POP3

(C) SMTP, MIME

(D) IMAP, SMTP

Ans: SEND → SMTP Retrieve → POP3 IMAP → Retrieve

MIME → Extension.

Que: Which of the following protocols is not used to resolve one form of address to another one?

(A) DNS

(B) ARP

(C) DHCP

(D) RARP

Ans: → DHCP are used to get the Ip address but not to resolve one from to another.

Que: Which of the following is/are examples of stateful application layer protocols?

(i) HTTP

(ii) FTP

(iii) TCP

(iv) POP3

Ans: HTTP not

FTP stateful

TCP ✗

POP ✓

(A) (i) and (ii) only

(B) (ii) and (iii) only

(C) (iii) and (iv) only

(D) (iv) only.

Que: Which of the following is not correct about HTTP Cookies?

(A) A Cookies is a piece of code that has the potential to compromise the security of an Internet user.

(B) A cookie gains entry to the user work area through an HTTP header.

(C) A cookie has an expiry date and time.

(D) Cookies can be used to track the browsing pattern of a user at a particular site.

Ans:-

B → Cookie stored by httpheader

C → Yes, they have TTL

D → Yes they are used for that.

Que: Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

(A) HTTP GET request, DNS Query, TCP SYN

(B) DNS Query, HTTP GET req, TCP SYN

(C) DNS Query, TCP SYN, HTTP GET Request

(D) TCP SYN, DNS Query, HTTP GET request -

Ans: First we need → DNS Query to resolve hostname.

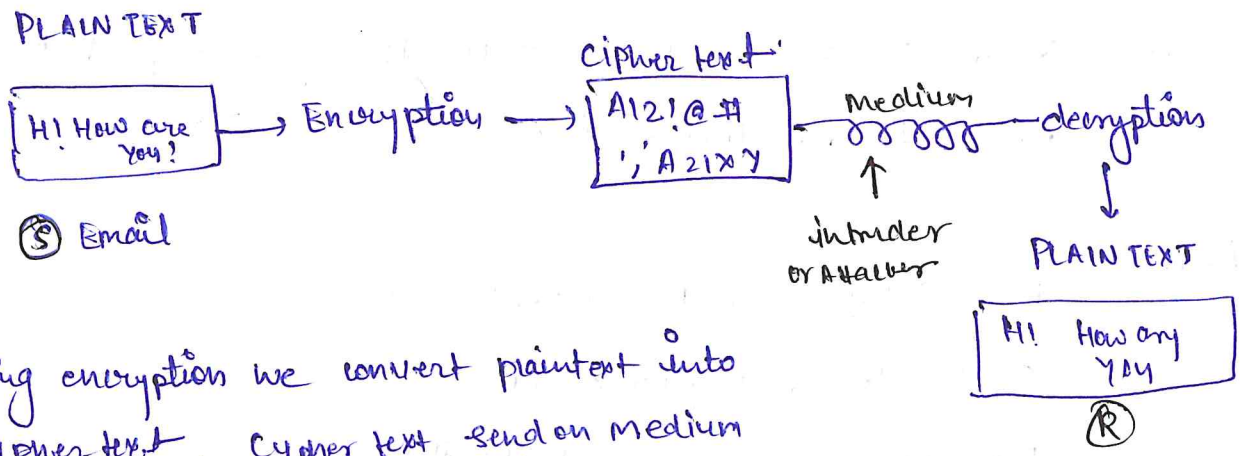
Second → TCP SYN to establish a connection.

GET → To get the page.

Network security and Encryption [Alan Turing during WW-II]

Prime fact Group theory

Classical encryption:



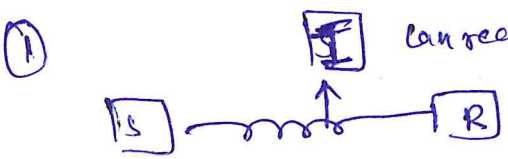
→ During encryption we convert plaintext into cypher text. Cypher text send on medium

→ If intruder get the message. It won't understand anything.

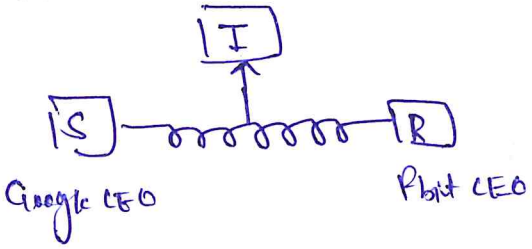
→ During decryption cypher text converted to plain text which can are used by receiver.

Passive vs Active attacker

→ Attacker can read the message but not modify it is called passive attacker



(2) Traffic pattern Analysis

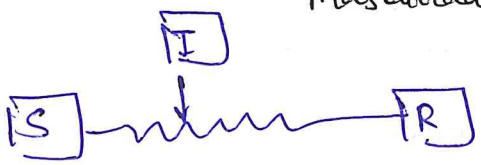


→ Passive Attacker.

* Intruder can track how many E-mails are send from google CEO to Fbit CEO. It will start doing data analysis. Many important information can be done.

Active Attacker :-

(1) Masquerade / pretending

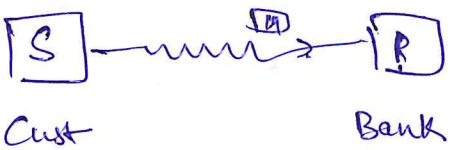


→ If Intruder can be pretend like a sender then receiver will be misguided.

→ In online banking it is dangerous.

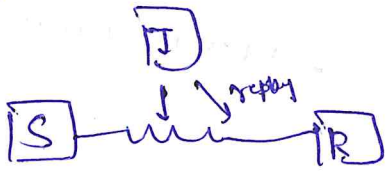
(2) Message modification → Attacker can modify the message of sender.

(3) Repudiation (deny) →



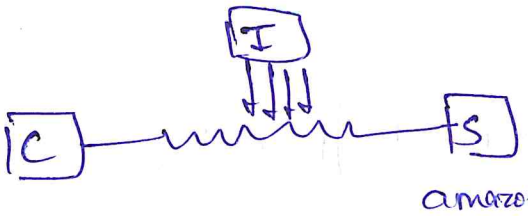
→ If someone deny that this message was not sent by them.

(4) Replay →



Attacker replaying the same packet that is captured from sender to receiver.

(5) Denial of service:



To down the server of amazon, Intruder is pushing so many requests acting as client because of so many requests the server is down.

The authentic user will not able to take the service.

Services expected by a security system:

(1) Authentication → Receiver should know whether packet is coming from authentic sender.

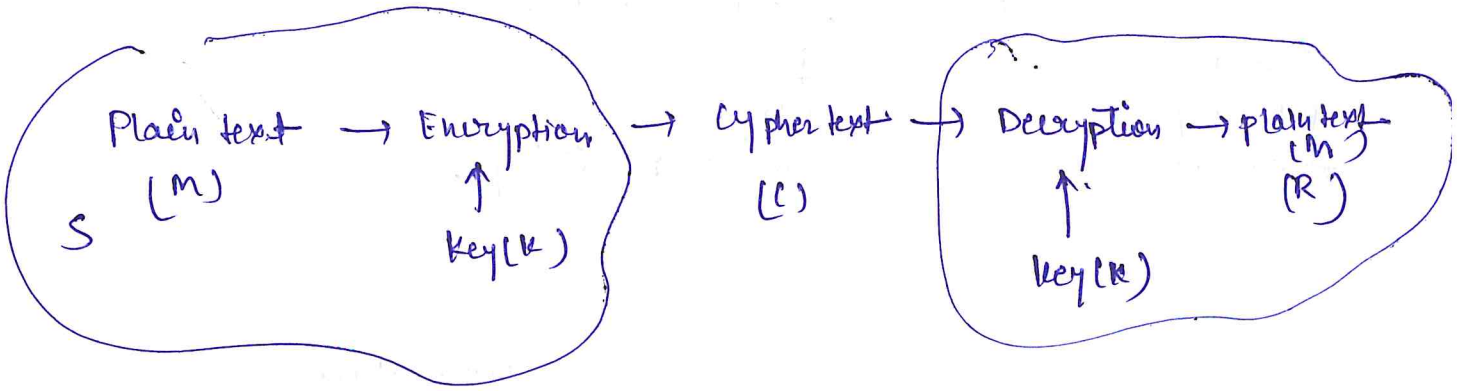
(2) Data integrity → We want data must not tempered between sender and receiver.

(3) Access-control → Gmail login, pass, OTP etc.

(4) Confidentality of data → I don't want anybody to read the information in between.

(5) Non-repudiation → The person can not deny its usage therfore OTP etc are used.

Cryptography: Symmetric



→ We are using same key for encryption and decryption therefore it is called Symmetric cryptography.

Eg: DES, AES etc.

Challenge ⇒ How sender and receiver can agree on the same key. [key exchange]

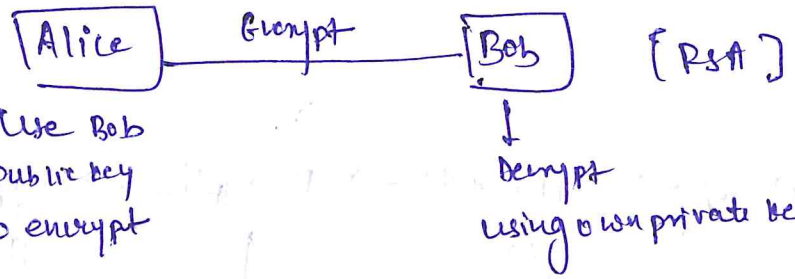
Cryptography: Assymetric key cryptography {RSA algorithm}

→ In Assymetric key cryptography we use two keys: Public key and private key

→ We shared the public key to everyone. Sender and receiver public keys are available and private keys are not known by anyone outside.

Public key

Private key → Not share.



Modulo Arithmetic :—

Number theory

[Mathematics required]

$a \equiv b \pmod{n}$ $\Rightarrow$ a R b under mod n

$\Rightarrow$ a is congruent to b mod n

e.g. $38 \equiv 14 \bmod 12$

$\Downarrow$

This equal to $(a-b)/n$ $\Leftrightarrow$ $ak + n$ (a−b divisible by n)

e.g. $38 \equiv 2 \bmod 12$ ✓ Yes

e.g. $-8 \equiv 7 \bmod 5$

$-8-7/5 = -5$ ✓ [Yes divisible]

Properties:-

(1) st $a \equiv b \bmod n$ and $c \equiv d \bmod n$.

$\Rightarrow (a+c) \equiv (b+d) \bmod n$

Proof: $a = nK + b$

$c = nl + d$

$(a+c) = n(K+l) + (b+d)$

(2) nd $a_1 \equiv b_1 \bmod n$ and $a_2 \equiv b_2 \bmod n$

$\Rightarrow a_1 \cdot a_2 = b_1 \cdot b_2 \bmod n$

$\Rightarrow a_1^K \equiv b_1^K \bmod n$ $\forall$ non-negative integers K.

(3) rd $p(x)$: polynomial function

$\Rightarrow p(a) \equiv p(b) \bmod n$

* $(a \equiv b \bmod n) \not\Rightarrow (K^a \equiv K^b \bmod n)$

Not always

(4) $\{a \equiv (b+c) \bmod n\} \Rightarrow \{a \equiv (b \bmod n + c \bmod n) \bmod n\}$

$\{a \equiv (b \cdot c) \bmod n\} \rightarrow \{a \equiv (b \bmod n \cdot c \bmod n) \bmod n\}$

Modular multiplicative Inverse :

$ax \equiv 1 \pmod n \iff x \equiv a^{-1} \pmod n$

a and x multiplicative Inverses of each other under modulo n.

Primes and co-primes (relative-primes) [Number theory]

$\Sigma_{13,5,17,11}$

$\gcd(x,y) = 1$

$\gcd(12,17) = 1$

So, they are co-primes.

* If x and y are co-primes then $\exists\ z \in I$ s.t

$xz \equiv 1 \pmod y$

x and z are modulo multiplative Inverse of each other

eg: $x = 12, y = 17$

$\Rightarrow xz \equiv 1 \bmod y$

$\Rightarrow 12x \begin{pmatrix} 1 \\ 2 \\ 3 \\ \vdots \\ 10 \end{pmatrix} \equiv 1 \bmod 17 \qquad z = 10$

→ Try for 1, 2, ...

Euler Totient function :—

$\phi(n)$ = Number of +ve integers upto n that are co-primes to n.

Eg: $\phi(9) = 6$

1, 2, 3, 4, 5, 6, 7, 8, 9

Total = 6

* If n and m are co-prime then

$\phi(m)\,\phi(n) = \phi(n \cdot m)$

* If $n = p_1^{k_1} \cdot p_2^{k_2} \cdot p_3^{k_3} \cdots p_r^{k_r}$

then

$$\phi(n) = n\left[1-\frac{1}{p_1}\right]\left[1-\frac{1}{p_2}\right]\cdots\left[1-\frac{1}{p_r}\right]$$

↑ Euler product formula

Eg: $n = 36$

$= 2^2 \cdot 3^2$

$\phi(n) = 36\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right) = 36 \cdot \frac{1}{2} \cdot \frac{2}{3} = 12$

Euler's Theorem : [co-primes]

if $n \in I^+$ & a, n are co-prime

then $a^{\phi(n)} \equiv 1 \pmod{n}$

e.g: $n = 165$

$\phi(n) = \phi(165) = \phi(15)\,\phi(11)$

$= 15\left(1-\frac{1}{3}\right)\left(1-\frac{1}{5}\right) \cdot 11\left(1-\frac{1}{11}\right)$

$\phi(n) = 8 \cdot 10 = 80$

$8^{80} \equiv 1 \bmod 165$

Fermat's little theorem: Simple extension

→ If $n \in I^+$ & a, n are co-primes

then $a^{\phi(n)} + 1 \equiv a \pmod{n}$

$$\begin{cases} a^{\phi(n)} \equiv 1 \pmod{n} & \rightarrow \text{Euler} \checkmark \\ a \equiv a \pmod{n} & \rightarrow \text{Trivial} \checkmark \end{cases}$$

Another Extension:

If $n \in I^+$ & a, n are co-prime & $b \equiv 1 \pmod{\phi(n)}$

then $a^b \equiv a \pmod{n}$

Proof: $b \equiv 1 \pmod{\phi(n)}$

$b = k \cdot \phi(n) + 1$.

$a^b = a^{k\phi(n)+1} = (a^{\phi(n)})^k \cdot a$

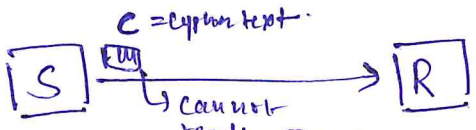
$a^{\phi(n)^k} \equiv 1^k \equiv 1 \pmod{}$

$a \equiv a \pmod{n}$

↳ Trivial

RSA (Rivest, Shamir, Adelman)

[In RSA sender encrypt with receiver public key and receiver decrypt with own private key]



K_u^R = R's public key

K_{pr}^R = R's private key

$K_u^R = (e, n)$

P = plain text

$C = P^e \bmod n$

$0 \le P < n$

$K_{pr}^R = (d, n)$

$P = C^d \bmod n$

$\Rightarrow P = (P^e \bmod n)^d = P^{ed} \bmod n$

$P < n$ → means [1234 | 1234] → If we have bigger plain text we divide in blocks for sending.

* R's public key known to everyone = K_u^R
* R's private key only known to R.
*

~~Operation at R =~~

* We want n, e, d to satisfy $P^{ed} \bmod n = P$.
  - At receiver side to again get plain text
* We want to pick e, d s.t

$$P = P^{ed} \bmod n$$

$$ed = 1 \bmod \phi(n)$$ → Theorem.

(1) Given (e, n) computing d must be very ~~too~~ hard.

(e, n) → K_u^R

⇓

Computing $\phi(n)$ must be hard

n, e, d in RSA

① P_1, P_2 be large prime numbers.

$n = P_1 \cdot P_2$. ⇒ n is very large [256 - bit numbers]

$\phi(n) = \phi(P_1) \cdot \phi(P_2) = (P_1 - 1)(P_2 - 1)$

② Pick $1 < e < \phi(n)$ s.t $\gcd(e, \phi(n)) = 1$ ($e, \phi(n)$ are co-prime)

⇒ $\exists d \in I^+$ s.t $e \cdot d \equiv 1 \bmod \phi(n)$

(modulo multi Inverse)

*

Eg. $P_1 = 53$ $P_2 = 61$

$n = 53 \times 61 = 3233$

$\phi(n) = \phi(53)\,\phi(61) = 52 \times 60 = 3120$

$e = 17$, pick some e

$\gcd(17, 3120) = 1$.

$e \cdot d \equiv 1 \bmod \phi(n)$

17 ↓ 3120 ↓

Then, $d = 2753$

$K_u R = (17, 3233)$ (e, n)

$K_R R = (2753, 3233)$ (d, n)

Residue:

$a \equiv b \bmod n \iff b$ is residue of $a \bmod n$

(residue: remainder)

e.g. $38 \equiv 14 \bmod 12 \iff 14$ is residue of $38 \bmod 12$

Residue-class / Congruence class

Defn: $\bar{a}_n = \{ \ldots, a-2n, a-n, a, a+n, a+2n, a+3n \ldots \}$

↳ Set of all integers that are $\equiv a \bmod n$

e.g. The residue class of $x^2 \bmod 6$ are

$0^2 \bmod 6 = 0$

$1^2 \bmod 6 = 1$

$2^2 \bmod 6 = 4$

$3^2 \bmod 6 = 3$

$4^2 \bmod 6 = 4$

$5^2 \bmod 6 = 1$

The residue here → $0_6, 1_6, 3_6, \bar{4}_6$

↳ You will get only these residue pattern

Primitive root of prime :—

P : prime number

g is set to primitive root of n if $g^0, g^1, g^2 \ldots$ (exponent of g)

includes all congruence-classes of P $\{\bar{1}_p, \bar{2}_p, \bar{3}_p, \bar{4}_p, \bar{5}_p \ldots\}$

e.g. $P=7 \quad g=3$

$3^0 \bmod 7 = 1$
$3^1 \bmod 7 = 3$
$3^2 \bmod 7 = 2$
$3^3 \bmod 7 = 6$

$3^4 \bmod 7 = 4$
$3^5 \bmod 7 = 5$
$3^6 \bmod 7 = 1$
⋮

3 is the primitive root of 7

→ 7 has multiple primitive roots not that one.

e.g. $P=7 \quad g=5$

$5^0 \bmod 7 = 1$
$5^1 \bmod 7 = 5$
$5^2 \bmod 7 = 4$
$5^3 \bmod 7 = 6$

$5^4 \bmod 7 = 2$
$5^5 \bmod 7 = 3$
$5^6 \bmod 7 = 1$
$5^7 \bmod 7 = 5$ } repeat

for $P=7$ 3, 5 primitive roots.

Fermat's little theorem :

$$g^{p-1} \equiv \text{mod } p \quad \text{as} \quad \phi(p) = p-1.$$

* Powers formed by g can't have cycle longer than $p-1$ if cycle-length $= p-1$ then g is a primitive root.

Alternative defn with cycle length

Multiplicative order :

Coprimes a, n, the multiplicative order of a mod n is the smallest $k \in I^+$ s.t

$$a^k \equiv 1 \text{ mod } n.$$

Generalised defn : { a is primitive root of n iff multiplicative order of $a = \phi(n)$ }

Let n is prime $\phi(n) = n-1$

* The concept of multiplicative order and cycle length both are related.

n	g(n)	
2	1	
3	2	
4	3	
5	2, 3	$\rightarrow 5^1$
6	5	$\rightarrow 2 \cdot 3^1$
7	3, 5	$\rightarrow 7^1$
9	2, 5	$\rightarrow 3^2$
10	3, 7	$\rightarrow 2 \cdot 5^1$
11	2, 6, 7, 8	$\rightarrow 11^1$
13	2, 6, 7, 11	$\rightarrow 13^1$

Except 1, 2 and 4 all the other integers with primitive roots are of the form p^k or $2p^k$ where p is an odd prime.

Discrete Logarithm problem :-

Given a, b, p find x s.t $a^x \equiv b \bmod p$

↳ Large prime (256 bit.)

* given a, x and p computing b is easy. [Try for all b]

* given a, b and p computing x is very hard.

↳ computationally hard

One-way funtions:

$f(x) = b$ → easy

→ easy to compute

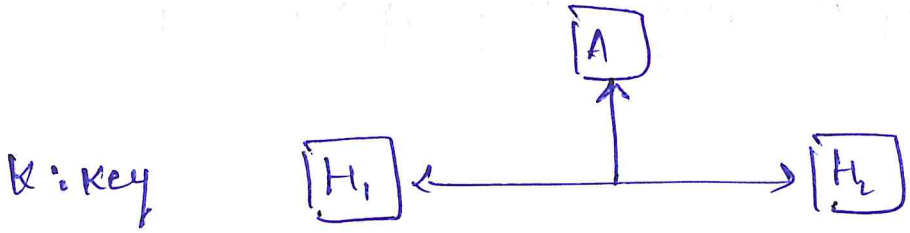
a, ϕ

$g(b) = x$ → v. hard to compute

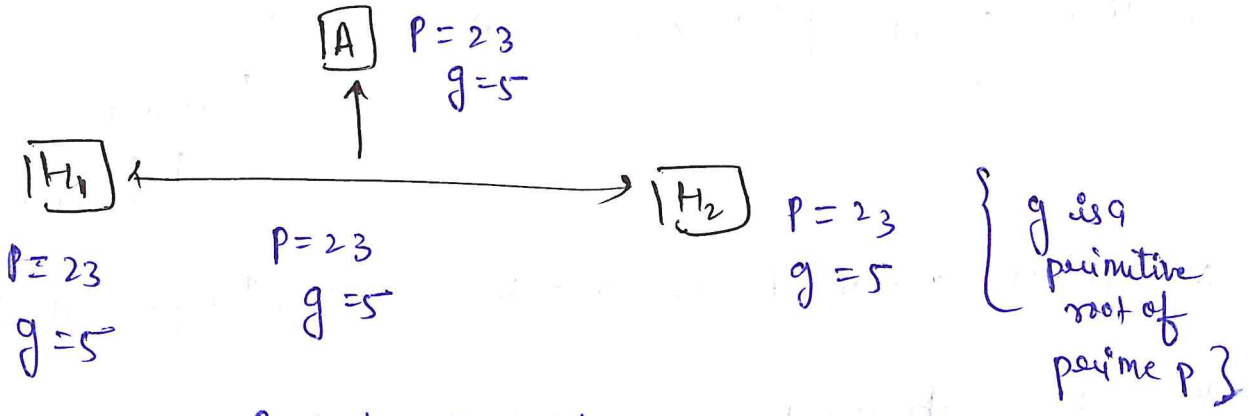
a, p

Diffie-Hellman key exchange :- (SSL/TL)

- Symmetric Encryption : key exchange.



Step-1 (1)



P and g known to everyone

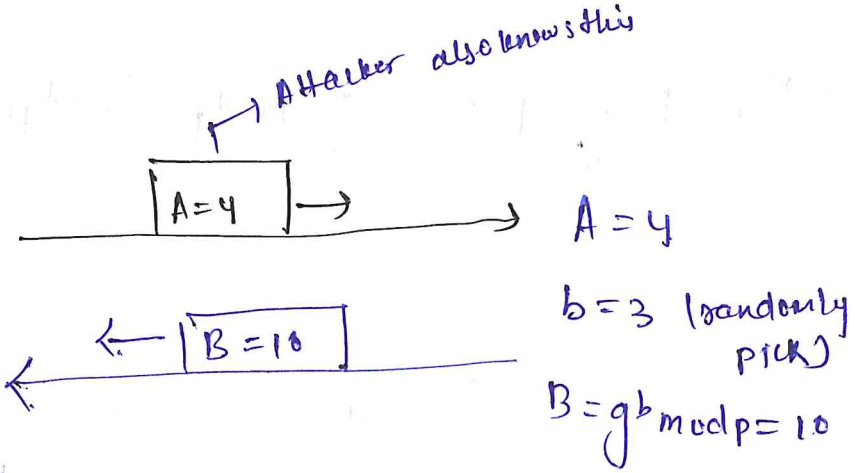
(2) ~~This is secret~~

$a = 4$ (randomly)

$A = g^a \bmod p = 4$

Computes this

$B = 10$



* a is the secret known to H1
* b is the secret known to H2

Host A knows

$A = 4$

$a = 4$

$B = 10$

$p = 23$

$g = 5$

$K = B^a \bmod p$

↕

Attacker can't compute this

Attacker knows

$p = 23$

$g = 5$

$A = 4$

$B = 10$

↓

Attack can't figure out a, b

Host B knows

$p = 23$

$g = 5$

$b = 3$

$A = 4$

$B = 10$

$K = A^b \bmod p$

↓

Attacker can't compute

finding exp is hard

$g^{(a)} = A \bmod p$

$B^a \bmod p = A^b \bmod p$ → equal

Proof: $B^a \bmod p = (g^b \bmod p)^a \bmod p = (g^b)^a \bmod p$

$= (g^a)^b \bmod p$

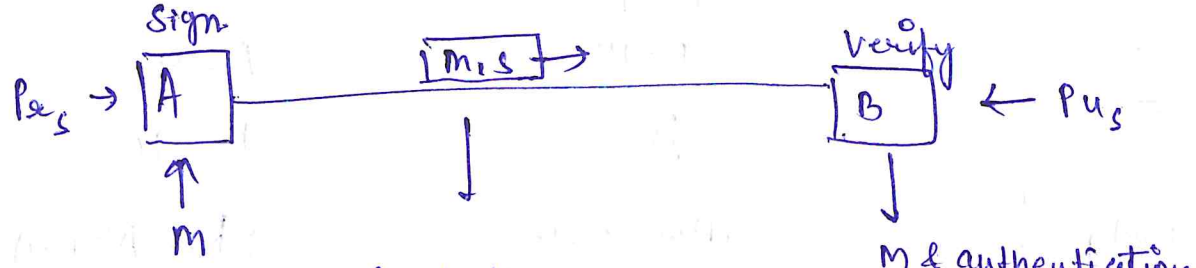
$= (g^a \bmod p)^b \bmod p$

$= A^b \bmod p$

eq used.

Digital signature (authenticity-verification)

- Using Public key Cryptography.
- Proof of document authenticity we use digital signature.



A is the one who signed the signature. that B should know.

* Sender Encrypt with its own private key
* Public key of sender known to everyone.
* Private key of sender is only known to sender.

→ We encrypt using private key of sender

$$S = \text{encrypt}\ (M, Pr_S)$$

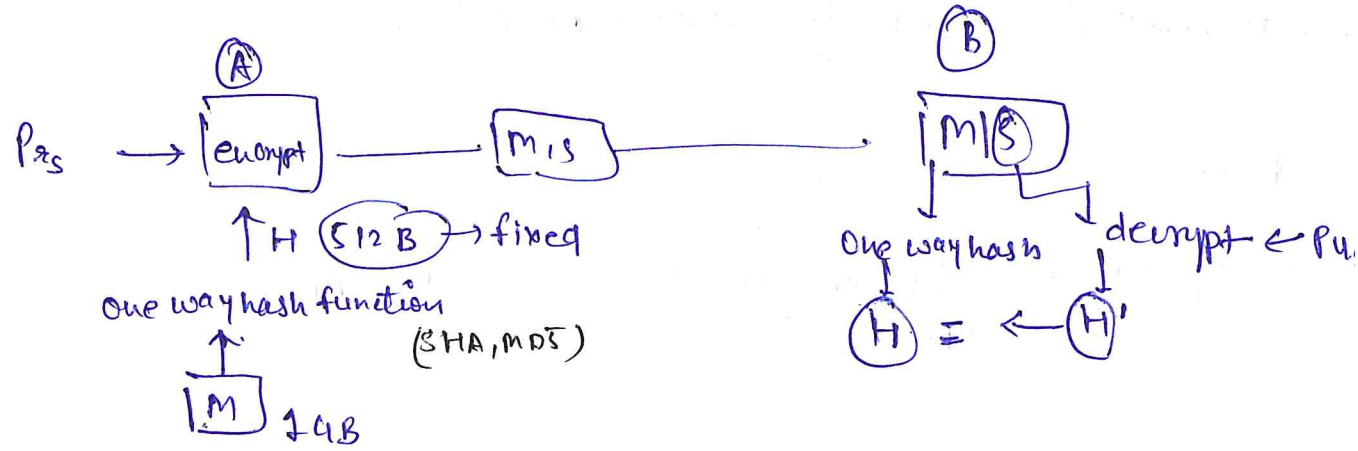
→ B has public key of sender, After decrypting with public key of A sender got to know this message is send by A.

$$M = M' = \text{decrypt}\ (S, Pu_S)$$

* Problem → Here problem is we are sending bigger amount of data. Encrypting it takes much time.

To Solve this issue:

→ We will compute the one way hash function using message. It will generate 512B hash value.



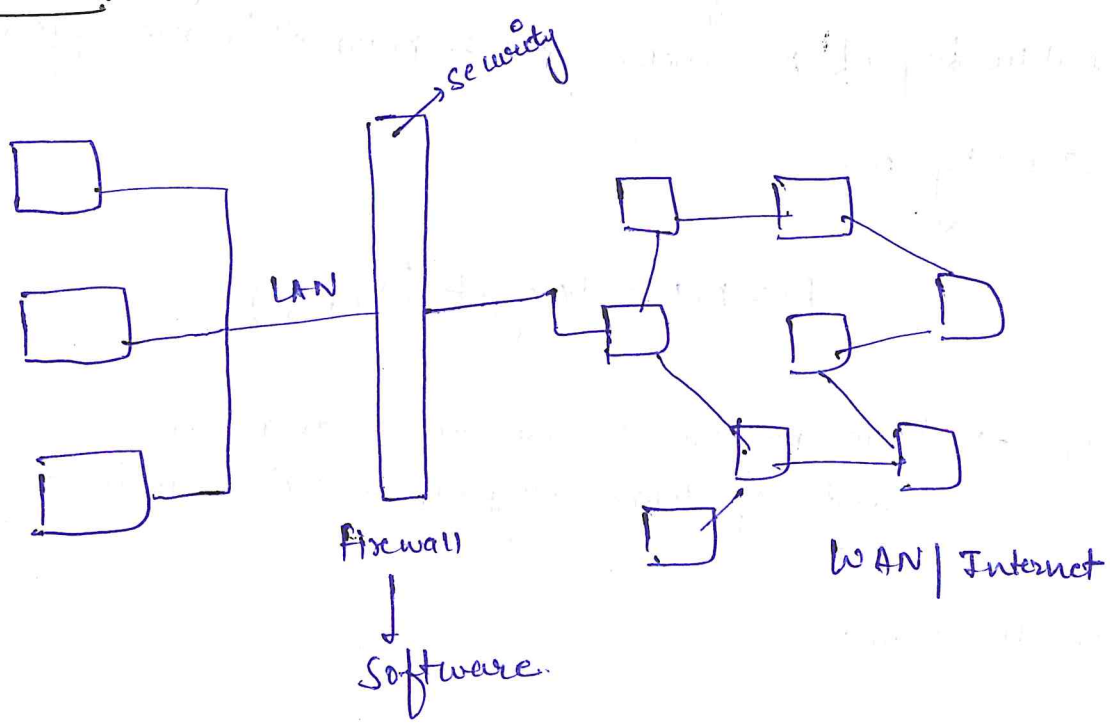
* We encrypt hash value, so it will become small size.

* We decrypt using public key

* We compute hash on m

If $H = H'$ then it is verified.

Firewalls:



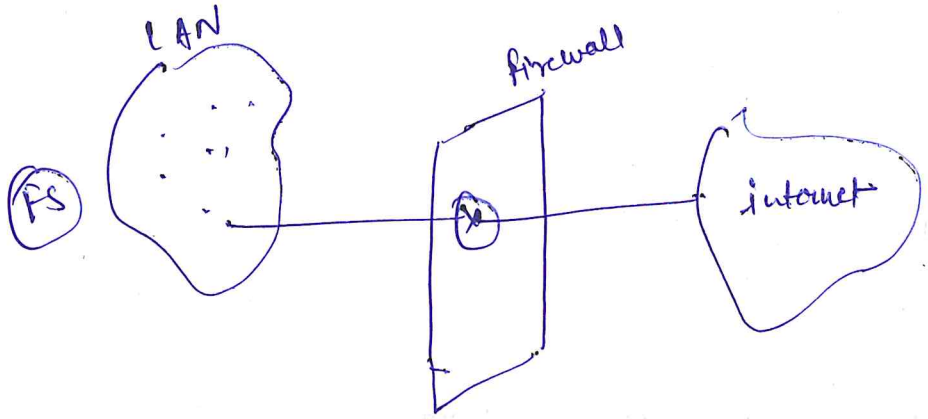
→ To block unnecessary packets to go from LAN to outside and to block unnecessary packets from outside network to LAN.

→ Firewall is used for security mechanism

Packet filter firewall:

→ operates @ IP and TCP layer ← NOTAL
(N/w) TCP

→ Filter based on IP and port address.



e.g: Block port 21 ←

↓

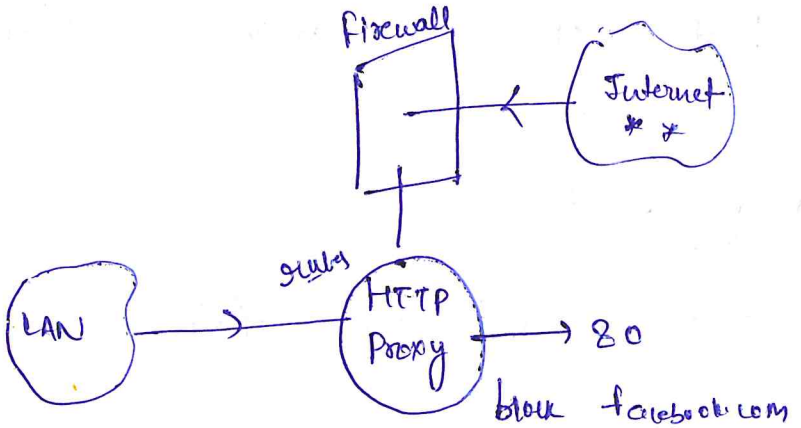
→ No computer from outside can access file system of LAN. We are blocking port 21

→ Block 121.62.1.*

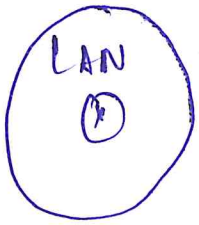
No packet from LAN can go to 121.62.1.* → Hacked sites / ~~fb~~ social network fb etc

Application Layer | proxy firewalls

→ It works on Application layer.



→ Any flow which is HTTP packet will first recive by HTTP proxy.



→ HTTP header checked by HTTP proxy. The blocked websites we can not access further.

External proxy servers:

The websites are blocked by local proxy server we can use external proxy server to access the website that are blocked by local proxy server.

Solved problems:

Que: In RSA cryptosystem, the value of the public modulus parameter n is 3007. If it is also known that $\phi(n) = 2880$, where $\phi()$ denotes Eulers totient function, then the prime factors of n which is greater than 50 is ____?

Ans: $n = 3007 = p \cdot q$ —①

$\phi(n) = 2880 = (p-1)(q-1)$ —②

$$2880 = pq - p - q + 1$$

$$2880 = 3007 - p - q + 1$$

$$p + q = 128 \quad \text{—(3)}$$

$$p = \frac{3007}{q}.$$

$$3007 + q^2 = 128q$$

$$q = 31, (97) \leftarrow \text{greater than 50}$$

Que A sender S sends a message m to the receiver R, which is digitally signed by S with its private key. In this scenario, one or more of the following security violations can take place.

Using Birthday Attack, some fraudulent message can be generated which has the same hash value (due to hash collisions) and digital signature as the original message.

(I) S can launch birthday attack to replace m with the fraudlent message.

(II) A third party attacker can launch a birthday attack to replace m with a fraudlent message.

(III) R can launch a birthday attack to replace m with a fraudulent message.

(A) I and II only

(B) I only

(C) II only

(D) (II) and (III) only.

Ans:

S —— m, sign. ——→ R

* In digital signatures we pass message m and by hash function we generate hash value.

m —hash→ (H)

m' —hash→ (H)

→ (1) Yes can send message m' by using same hash value.

→ (2) Not possible

→ 3 Not possible

Que: In RSA cryptosystem a particular A uses two prime numbers $p=13$ and $q=17$ to generate her public and private keys. If a public key of A is 35. Then the private key of A is ____.

Ans:

$p \cdot q = n = 13 \times 17$

$\phi(n) = 12 \times 16 = 192$

e, n, d

$e = 35$: Public

$d = Pr = ?$

$\gcd(35, 192) = 1 \qquad 1 < e < \phi(n)$

$1 < d < \phi(n)$

$\Rightarrow ed \bmod \phi(n) = 1$

$\Rightarrow 35 \cdot d \bmod 192 = 1$

$d = 11$

Que. Consider that B wants to send a message m that is digitally signed to A. Let the pair of private and public keys for A and B be denoted K_x^- and K_x^+ for $x = A, B$, respectively. Let $K_x(M)$ represent the operation of encrypting m with a key K_x and $H(m)$ represent the message digest. Which one of the following indicates the CORRECT way of sending the message m along with the digital signature to A?

A. $\{m, k_B^+ H(m))\}$

B. $\{m, k_B^- (H(m))\}$

C. $\{m, k_A^- (H(m))\}$

D. $\{m, k_A^+ (m)\}$

Ans:- → private
+ → public

$m \xrightarrow{h}$ (H) → message digest

B wants to send a message.

B ⟶ A

$\downarrow [m, k_B^- H(m)]$ ⟶ Ans

We use B private key

Que: Anarkalli digitally signs a message and sends it to salim. verification of the signature by salim requires.

(A) Anarkali's public key
(B) Salim's public key
(C) Salim's private key
(D) Anarkali's private key

→ In authentication reciever decrypts with sender's public key.

A ⟶ S

P_{r-A} ↓ Public key of Anarkali